Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Let A and B be two sets then $(A \cup B)' \cup (A' \cap B)$ is equal to

(A) A'

(B) A

(C) B '

(D) None of these

Q.2

At a certain conference of 100 people, there are 29 Indian women and 23 Indian men. Of these Indian people 4 are doctors and 24 are either men or doctors. There are no foreign doctors. How many foreigners and women doctors are attending the conference?

(A) 48,1

(B) 34,3

(C) 46,4

(D) 42,2

Q.3

Let A and B are two sets in a universal set U. Then which of these is/are correct?

(A) $A - B = A' - B'$

(B) $A - (A - B) = A \cap B'$

(C) $A - B = A' \cap B'$

(D) $A \cup B = (A - B) \cup (B - A) \cup (A \cap B)$

Q.4

Each student in a class of 40 , studies at least one of the subjects English, Mathematics and Economics. 16 study English, 22 Economics and 26 Mathematics, 5 study English and Economics, 14 Mathematics and Economics and 2 study all the three subjects. The number of students who study English and Mathematics but not Economics is

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(A) 7

(B) 5

(C) 10

(D) 4

Q.5

In a class of 60 students, 23 play Hockey 15 Play Basket-ball and 20 play cricket. 7 play Hockey and Basket-ball, 5 play cricket and Basket-ball, 4 play Hockey and Cricket and 15 students do not play any of these games. Then

(A) 4 play Hockey, Basket-ball and Cricket

(B) 20 play Hockey but not Cricket

(C) 1 plays Hockey and Cricket but not Basket-ball

(D) All above are correct

Q.6

The number of students who take both the subjects mathematics and chemistry is 30. This represents 10% of the enrolment in mathematics and 12% of the enrolment in
chemistry. How many students take at least one of these two subjects?

(A) 520

(B) 490

(C) 560

(D) 480

Q.7

The value of $(\mathrm{A} \cup \mathrm{B} \cup \mathrm{C}) \cap (\mathrm{A} \cap \mathrm{B}^c \cap \mathrm{C}^c)^c \cap \mathrm{C}^c$, is

(A) $\mathrm{B} \cap \mathrm{C}^c$

(B) $\mathrm{B}^c \cap \mathrm{C}^c$

(C) $\mathrm{B} \cap \mathrm{C}$

(D) $A \cap B \cap C$

Q.8

If $\mathrm{A}, \mathrm{B}, \mathrm{C}$ are three sets, then what is $\mathrm{A} - (\mathrm{B} - \mathrm{C})$
equal to?

(A) $A - (B \cap C)$

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(B) $(A - B) \cup C$

(C) $(A - B) \cup (A \cap C)$

(D) $(A - B) \cup (A - C)$

Q.9

Let $A = \{1, 2, 3\}$, then the relation $R = \{(1,1), (1,2), (2,1)\}$ on A is

(A) reflexive

(B) transitive

(C) symmetric

(D) none of these

Q.10

A survey of 500 television viewers produced the following information, 285 watch football, 195 watch hockey, 115 watch basket-ball, 45 watch football and basket ball, 70 watch football and hockey, 50 watch hockey and basket ball, 50 do not watch any of the three games. The number of viewers, who watch exactly one of the three games are

(A) 325

(B) 310

(C) 405

(D) 372

Q.11

A dinner party is to be fixed for a group of 100 persons. In this party, 50 persons do not prefer fish, 60 prefer chicken and 10 do not prefer either chicken or fish. The number of persons who prefer both fish and chicken is.

(A) 20

(B) 22

(C) 25

(D) None of these

Q.12

Which is the simplified representation of $(A' \cap B' \cap C) \cup (B \cap C) \cup (A \cap C)$ where A, B, C are subsets of set X ?

(A) A

(B) B

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(C) C

(D) $X \cap (A \cup B \cup C)$

Q.13

Which one of the following is correct

(A) $A \cup (B - C) = A \cap (B \cap C')$

(B) $A - (B \cup C) = (A \cap B') \cap C'$

(C) $A - (B \cap C) = (A \cap B') \cap C$

(D) $A \cap (B - C) = (A \cap B) \cap C$

Q.14

If $\mathrm{F(n)}$ denotes the set of all divisors of n except 1 , what is the least value of y satisfying $[F(20) \cap F(16)] \subseteq F(y)$?

(A) 1

(B) 4

(C) 2

(D) 8

Q.15

If μ is the universal set and P is a subset of μ, then what is $\mathrm{P} \cap \{(\mathrm{P} - \mu) \cup (\mu - \mathrm{P})\}$ equal to ?

(A) ϕ

(B) P'

(C) m

(D) P

Answers & Solutions

Q.1 Answer:

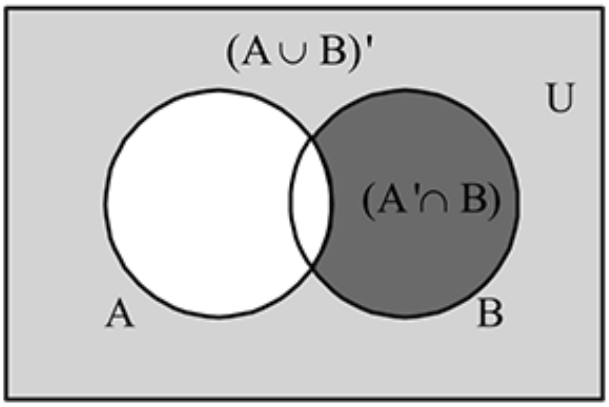
A'

Solution:

From Venn-Euler's Diagram.

$(A \cup B)' \cup (A' \cap B) = A$

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Q.2 Answer:

48,1

Solution:

See the following Venn diagram

$\mathrm{n(I)} = 29 + 23 = 52$

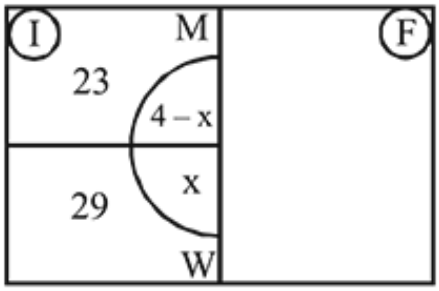
$\mathrm{n(F)} = 100 - 52 = 48$

$n(M \cup D) = n(M) + n(D) - n(M \cap D)$

$24 = 23 + 4 - n(M \cap D)$

$\therefore \mathrm{n(M \cap D)} = 3$

$\therefore \mathrm{n(W \cap D)} = 4 - 3 = 1$



Q.3 Answer:

$A \cup B = (A - B) \cup (B - A) \cup (A \cap B)$

Solution:

(a) $x \in A - B \Leftrightarrow x \in A$ **and** $x \notin B$

$\Leftrightarrow x \in A$ **and** $x \in B' \Leftrightarrow x \in A \cap B'$

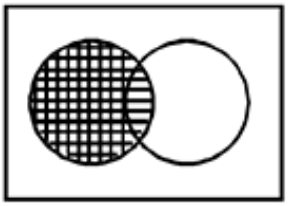
$\therefore \mathrm{A - B = A \cap B'} \ldots \mathrm{(i)}$

$x \in A$ **and** $x \in B'$

$\Leftrightarrow x \notin \mathrm{A'}$ **and** $x \in B' \Leftrightarrow x \in B'$ **and** $x \notin A'$

$\Leftrightarrow x \in B' - A'$

$\therefore \mathrm{A - B = B' - A'} \ldots$ **(ii)**



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Clearly (a) is not correct. Also from (i) (c) is not correct. Next let $x \in A - (A - B)$

$\Leftrightarrow x \in A$ **and** $x \notin A - B$

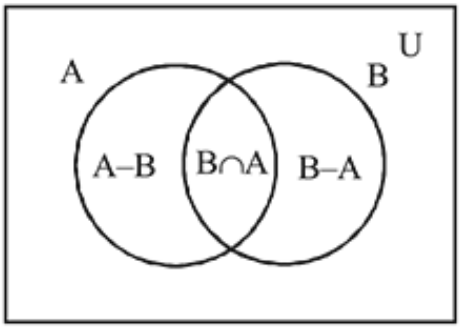
$\Leftrightarrow \mathrm{x} \in \mathrm{A}$ **and** $[\mathrm{x} \notin \mathrm{A}$ **or** $\mathrm{x} \in \mathrm{B}]$

$A - (A - B) = A \cap B$

$\Leftrightarrow [x \in A$ **and** $x \notin A]$ **or** $[x \in A$ **and** $x \in B]$ $\therefore \mathrm{A} - (\mathrm{A} - \mathrm{B}) \neq \phi \cup (\mathrm{A} \cap \mathrm{B}') = \mathrm{A} \cap \mathrm{B}'$

$\therefore$ **(b) is also incorrect The result** (d) **is correct as can be seen in the following Venn diagram**

$A \cup B = (A - B) \cup (A \cap B) \cup (B - A)$



Q.4 Answer:

5

Solution:

C **stands for set of students taking economics**

$\mathrm{a} + \mathrm{b} + \mathrm{c} + \mathrm{d} + \mathrm{e} + \mathrm{f} + \mathrm{g} = 40; \mathrm{a} + \mathrm{b} + \mathrm{d} + \mathrm{g} = 16$

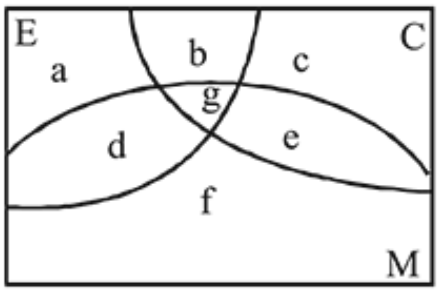
$\mathrm{b} + \mathrm{c} + \mathrm{e} + \mathrm{g} = 22; \mathrm{d} + \mathrm{e} + \mathrm{f} + \mathrm{g} = 26$

$\mathrm{b} + \mathrm{g} = 5; \mathrm{e} + \mathrm{g} = 14; \mathrm{g} = 2$

Go by backward substitution $\mathrm{e} = 12,\ \mathrm{b} = 3,\ \mathrm{d} + \mathrm{f} = 12, \mathrm{c} + \mathrm{e} = 17 \Rightarrow \mathrm{c} = 5$

$\mathrm{a} + \mathrm{d} = 11$

$a + d + f = 18 \Rightarrow f = 7 \quad \therefore d = 12 - 7 = 5$



Q.5 Answer:

1 plays Hockey and Cricket but not Basket-ball

Solution:

$\mathrm{a} + \mathrm{e} + \mathrm{f} + \mathrm{g} = 23$

$\mathrm{b} + \mathrm{d} + \mathrm{f} + \mathrm{g} = 15$

$\mathrm{c} + \mathrm{d} + \mathrm{e} + \mathrm{g} = 20$

$\mathrm{f} + \mathrm{g} = 7; \quad \mathrm{d} + \mathrm{g} = 5$

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$e + g = 4$

$a + b + c + d + e + f + g = 60 - 15 = 45$

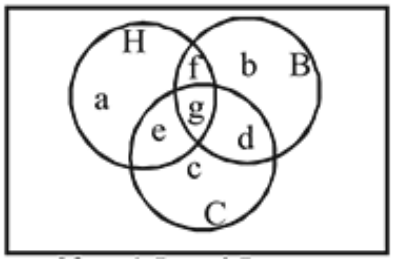
By substitutions, $a + e = 16,\ b + d = 8,\ b + f = 10, c + e = 15, c + d = 16$

Also, $b + c + d = 22$

$a + c + e = 30, a + b + f = 25$

From these, we get $b = 6, a = 15, c = 14 e = 1,\ d = 2, f = 4$ **and** $g = 3$

Clearly(a) is not correct for (b) $a + f = 19 \Rightarrow$ **(b)is incorrect for** $(c) e = 1 \Rightarrow (c)$ **is correct**



Q.6 Answer:

520

Solution:

Let the number of students who take only Math be x **and only Chemistry be** y**.**

So, from the Venn diagram, we have total number of students who take Math $= x + 30$ **and take Chemistry** $= y + 30$**. According to question, we have**

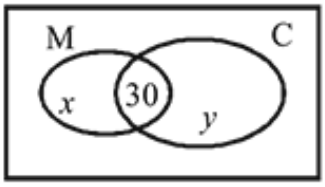
$$30 = \frac{10}{100}(x + 30)$$

$$\Rightarrow x = 270 \text{ and}$$

$$30 = \frac{12}{100}(30 + y)$$

$\Rightarrow y \quad = 220$

$x + y + 30 = 270 + 220 + 30 = 520$



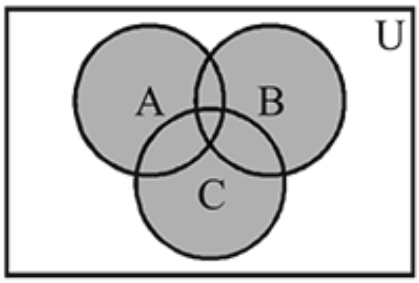
Q.7 Answer:

$B^c \cap C^c$

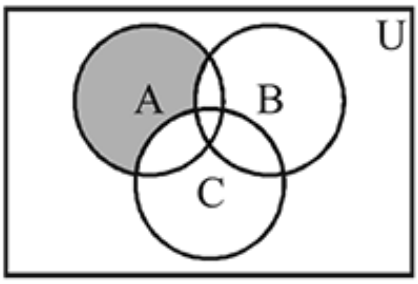
Solution:

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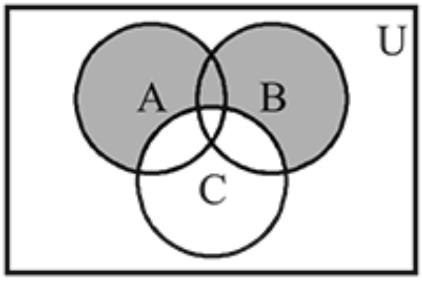
(a)



(i) $A \cup B \cup C$



(ii) $(A \cap B^c \cap C^c)$



(iii) C^c

From Fig. (i), (ii) and (iii), we get $(A \cup B \cup C) \cap (A \cap B^c \cap C^c)^c \cap C^c = (B^c \cap C^c)$

Q.8 Answer:

$(A - B) \cup (A \cap C)$

Solution:

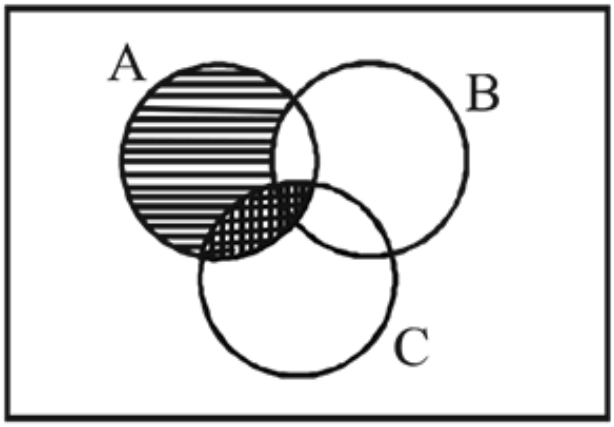
Following venn diagram shows the relation $A - (B - C)$

In the above venn diagram, horizontal lines shows

$(A - B)$ **and vertical lines show** $(A \cap C)$

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$\therefore (A - B) \cup (A \cap C) = A - (B - C)$



Q.9 Answer:

symmetric

Solution:

R is not reflexive as $(2,2) \notin R, (3,3) \notin R$

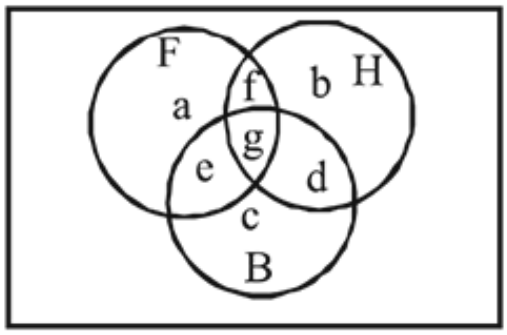
Also, R is not transitive as $(2,1) \in R$ and $(1,2) \in R$ but $(2,2) \notin R$.

However R is symmetric.

Q.10 Answer:

325

Solution:



$c + d + e + f = 115, e + g = 45, f + g = 70, \ d + g = 50$

$a + b + c + d + e + f + g = 500 - 50 = 450$

We obtain, $a + f = 240, b + d = 125, c + e = 65$

$a + e = 215, b + f = 145; b + c + d = 165$

$a + c + e = 255; a + b + f = 335$

Solving we get $b = 95, c = 40, a = 190, \ d = 30, e = 25, f = 50$ **and** $g = 20$

Desired quantity $= a + b + c = 325$

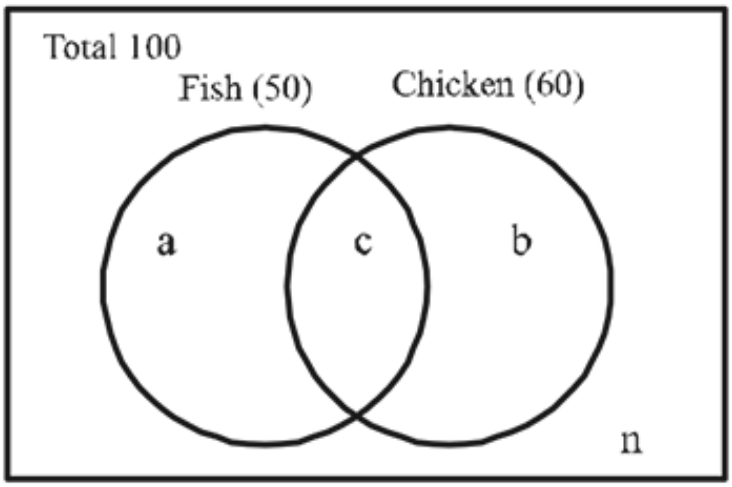
Q.11 Answer:

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20

Solution:

Total number of persons $= a + b + c + n = 100$



Do not prefer fish $b + n = 50$ **60 prefer chicken hence** $b + c = 60$ **Do not like fish and chicken is** $n = 10$

On solving these equations we will get $a = 30, b = 40, c = 20$

The number of persons who prefer both fish and chicken is $= c = 20$

Q.12 Answer:

C

Solution:

$A' \cap B' \cap C =$ **only** C

$\therefore \quad (\mathrm{A}' \cap \mathrm{B}' \cap \mathrm{C}) \cup (\mathrm{B} \cap \mathrm{C}) \cup (\mathrm{A} \cap \mathrm{C}) = \mathrm{C}$

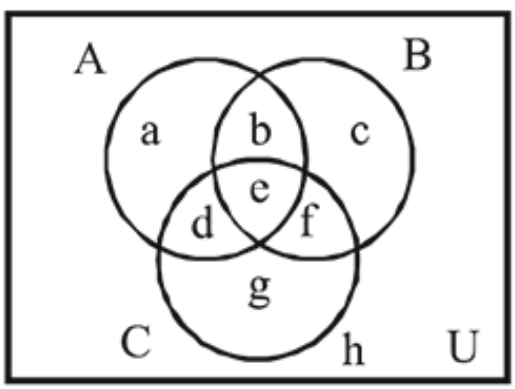
(By Venn Diagram)

Q.13 Answer:

$A - (B \cup C) = (A \cap B') \cap C'$

Solution:

Let a Venn-diagram be drawn taking three intersecting sets A, B **and** C **under a universal set** U**. This makes 8 regions a to** h **as shown.**



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A has regions $\mathrm{a}, \mathrm{b}, \mathrm{d}, \mathrm{e}$

B has regions b, c, e, f

C has regions d, e, f, g C' **has regions** $\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{h}$

B' **has regions** $\mathrm{a}, \mathrm{d}, \mathrm{g}, \mathrm{h}$

Statement (a) : $A \cup (B - C) = A \cap (B \cap C)$

$\mathrm{LHS} \equiv (\mathrm{a}, \mathrm{b}, \mathrm{e}, \mathrm{d}) \cup \mathrm{b}, \mathrm{c} \equiv \mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}, \mathrm{e}.$

$\mathrm{RHS} \equiv \mathrm{a}, \mathrm{b}, \mathrm{d}, \mathrm{e} \cap \mathrm{e}, \mathrm{f} \equiv \mathrm{e}$

So, statement (a) is not correct. Statement $(b) : A - (B \cup C) = (A \cap B') \cap C'$

$\mathrm{LHS} \equiv (\mathrm{a}, \mathrm{b}, \mathrm{d}, \mathrm{e}) - (\mathrm{b}, \mathrm{c}, \mathrm{d}, \mathrm{e}, \mathrm{f}, \mathrm{g}) \equiv \mathrm{a}$

$\mathrm{RHS} \equiv (\mathrm{a}, \mathrm{b}, \mathrm{d}, \mathrm{e} \cap \mathrm{a}, \mathrm{d}, \mathrm{g}, \mathrm{h}) \cap (\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{h}) \equiv \mathrm{a}$

So, statement (b) is correct. Correct statement is :

$A - (B \cup C) = (A \cap B') \cap C'$

Q.14 Answer:

4

Solution:

Given that $\mathrm{F}(\mathrm{n}) =$ **set of all divisors of** n **except**

$\therefore \quad \mathrm{F}(20) = \{2, 4, 5, 10, 20\}$

and $\mathrm{F}(16) = \{2, 4, 8, 16\}$

$\therefore \quad \mathrm{F}(20) \cap \mathrm{F}(16) = \{2, 4, 5, 10, 20\} \cap \{2, 4, 8\}$

$= \{2, 4\}$

Also, $\{F(20) \cap F(16)\} \subseteq F(y)$

So, least value of $\mathrm{y} = 4$

Q.15 Answer:

ϕ

Solution:

Since μ **is universal set and** $\mathrm{P} \subseteq \mu, \mathrm{P} - \mu = \phi$ **and** $\mu - \mathrm{P} = \mathrm{P}'$

So, $(\mathrm{P} - \mu) \cup (\mu - \mathrm{P}) = \phi \cup \mathrm{P}' = \mathrm{P}'$

Now, $\mathrm{P} \cap \{\mathrm{P} - \mu) \cup (\mu - \mathrm{P})\} = \mathrm{P} \cap \mathrm{P}' = \phi$

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